

Pipeline Performance Formulas

General Consideration About Pipeline:

- Time required to perform the 1st task = k cycles = $k * t_p$
- Time required to perform remaining $(n - 1)$ tasks = $(n - 1)$ cycles = $(n - 1) * t_p$
- For all n tasks = $(k + n - 1)$ cycles = $(k + n - 1) * t_p$

Non-Pipeline System:

- Time required to perform n tasks = $n * t_n$

Speed-up Ratio (S):

- $S = (n * t_n) / ((k + n - 1) * t_p)$
- When number of tasks increases ($n \gg k$), the ideal or maximum speed-up is:

$$S \text{ (ideal or max)} = t_n / t_p$$

Special Case:

- For 1 task, pipeline and non-pipeline both take the same time:

$$\Rightarrow t_n = k * t_p$$

$$\Rightarrow S \text{ (ideal)} = k$$

Cycle Time in Synchronous Pipeline:

- t_p = max of all segments delays
- t_p = max of all segments delays + buffer delay
- t_n = sum of all segment delays
- Maximum speed-up: $S \text{ (max)} = t_n / t_p$

Clock Rate and Cycle Time:

- $t_p = 1 / \text{pipeline clock rate}$
- $t_n = \text{no. of cycles for one task} * 1 \text{ cycle time}$

$$t_n = \text{CPI} * 1 \text{ cycle time}$$

Latency and Throughput:

- Non-pipeline latency = t_n
- Pipeline latency = t_p

In $(k + n - 1) * t_p$ time, number of tasks = n

- In 1 unit of time, number of tasks (Throughput) = $n / ((k + n - 1) * t_p)$
- In ideal conditions:

$$\text{Throughput} = 1 / t_p$$

Branch Instructions:

- If branch condition is evaluated in the i th stage, then the number of stalls due to branch instruction = $i - 1$